SUMS OF POWERS, TAXICABS, AND MAGIC SQUARES

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Abstract. We present a brief history of three interconnected problems: Diophantine equations involving sums of like powers, taxicab numbers and their generalizations, and magic squares.

1. Sums of Like Powers

We say that an equation is *Diophantine* if only integer solutions are of interest. Such equations have been studied since antiquity; indeed, the Pythagorean triple

$$3^2 + 4^2 = 5^2$$

has been known as a solution to $a^2 + b^2 = c^2$ since 1800 BC. Since then, we have proven there exist infinitely many such triples[1].

Not all Diophantine equations have solutions. For example, Fermat's last theorem[2] famously states that the Diophantine equation

$$a^k + b^k = c^k$$

has no solution consisting of positive integers for any integer power $k \geq 2$. Hence, we need not even consider the equation $a^3 + b^3 = c^3$. However, if we add a third term to the left-hand side, we obtain the equation

$$a^3 + b^3 + c^3 = d^3$$

which does have solutions, for example

$$3^3 + 4^3 + 5^3 = 6^3$$

for which reason 216 has sometimes been called "Plato's number." These observations led Euler to (falsely) conjecture in 1769 that if the equation

$$\sum_{i=1}^{n} a_i^k = b^k$$

has a solution consisting of positive integers, then $n \geq k$. It was not until 1966 when Lander and Parkin discovered a counterexample for the $k = 5$ case via a computer search,

$$27^5 + 84^5 + 110^5 + 133^5 = 144^5,$$

Date: July 3, 2026.

[1]In particular, there exist infinitely many *primitive* Pythagorean triples, meaning $\gcd(a, b, c) = 1$.

[2]Not proven until 1994 by Andrew Wiles.

which they published in a paper of merely two sentences [LP66].

The $k = 4$ case fell in 1988 when Elkies gave a counterexample, and within the same year, Frye found the smallest possible:

$$95800^4 + 217519^4 + 414560^4 = 422481^4.$$

With Euler's conjecture thoroughly debunked, Lander, Parkin, and Selfridge offered one of their own [LPS67]. In particular, that if the equation

$$\sum_{i=1}^{n} a_i^k = \sum_{j=1}^{m} b_j^k$$

has a solution of all positive integers with $a_i \neq b_j$ for all $1 \leq i \leq n$ and $1 \leq j \leq m$, then $m + n \geq k$. Observe that taking $m = 1$ yields something akin to Euler's conjecture, only with the bound loosened to $n \geq k - 1$. Though unproven, no counterexample to this conjecture is known as of writing.

2. Taxicab Numbers

2.1. Sums of cubes in two ways. The number 1729 has an interesting property that was made famous by the following anecdote from Hardy: when Ramanujan fell very ill, Hardy went to visit him in the hospital. Upon arriving, he remarked to Ramanujan that the number of the taxi in which he had come, 1729, was rather unremarkable. Immediately, Ramanujan objected:

> "No, it is a very interesting number; it is the smallest number expressible as a sum of two cubes in two different ways"

and indeed, we observe that

$$1729 = 1^3 + 12^3 = 9^3 + 10^3.$$

Following Ramanujan's example, we define the nth *taxicab number*, denoted $\text{Taxicab}(n)$, to be the smallest integer expressible as a sum of two cubes in n different ways. Hence, $1729 = \text{Taxicab}(2)$, and in the decades since Ramanujan, we have found several more,

$$\begin{aligned}
\text{Taxicab}(3) &= 87539319 \\
\text{Taxicab}(4) &= 6963472309248 \\
\text{Taxicab}(5) &= 48988659276962496 \\
\text{Taxicab}(6) &= 24153319581254312065344
\end{aligned}$$

which are also listed in A011541. Further, upper bounds are known for all taxicab numbers through Taxicab(19) as given in [Boy08].

TODO Generalized Taxicab numbers TODO Cabtaxi numbers TODO Open questions

2.2. **Differences of biquadrates in three ways.** Here *biquadrate* is simply an archaic term for the fourth power of a number. Using the search `cabtaxi 4 2 3 524288`, we find two numbers which can each be written as a difference of fourth powers in three distinct ways:

$$\begin{aligned} 4860992489864937000960 &= 335084^4 - 296668^4 \\ &= 264047^4 - 1169^4 \\ &= 265076^4 - 93436^4 \\ 25900232113758000049920 &= 401168^4 - 17228^4 \\ &= 421296^4 - 273588^4 \\ &= 415137^4 - 248289^4 \end{aligned}$$

Both of the above are among the 23 primitive solutions given in [CW07], with the latter also given earlier in [Zaj83].

2.3. **Sums of sursolids in two ways.** Again, *sursolid* is an archaic term for the fifth power of a number. Observe that a solution to the equation

$$a_1^5 + a_2^5 = b_1^5 + b_2^5$$

with a_1, a_2, b_1, b_2 distinct positive integers would be a counterexample to the Lander, Parkin, and Selfridge conjecture. Indeed, a solution to

$$a_1^k + a_2^k = b_1^k + b_2^k$$

for any $k \geq 5$ would disprove the conjecture; the conjecture implies that Taxicab$(k, 2, 2)$ does not exist for any $k \geq 5$. Hence, to disprove the conjecture, one needs only demonstrate Taxicab$(k, 2, 2)$ for some $k \geq 5$. Of course, this is easier said than done.

3. Magic Squares

3.1. **3x3 Magic Squares.** Consider the 3×3 matrix

$$[TODO]$$

which has the special property that the sums of each row, each column, and both diagonals are all equal to the same number. We call such a matrix a *magic square*. These have been long studied; in the late 1800s, Édouard Lucas showed that all 3×3 magic squares of distinct positive integers are parameterized by

$$\begin{bmatrix} c-b & c+(a+b) & c-a \\ c-(a-b) & c & c+(a-b) \\ c+a & c-(a+b) & c+b \end{bmatrix}$$

where a, b, and c are integers such that $0 < a < b < c-a$ and $b \neq 2a$.

TODO Impossibility of 3x3 cubes (and higher powers) TODO Open problem of 3x3 magic squares of squares TODO Open problem of 3x3 semimagic square of cubes

3.2. 4x4 Magic Squares. TODO 4x4 characterization TODO Open problem of 4x4 magic square of cubes

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